origin of the system of globular clusters in the milky way (massari et al)

https://ui.adsabs.harvard.edu/abs/2019A&A...630L...4M

**this study was referenced fairly often so i think the methods here are probably pretty sound**

- orbital parameters under closest consideration ○ z_max ○ apocentre radius ○ eccentricity ○ orbit circularity ○ age
    - ○ metallicity
- there is a distinction between metal rich and metal poor clusters between young galaxies on AMR graph ○ young metal rich galaxies have
        - smaller apocentres
        - smaller altitude from galactic plane (z_max)
        - lower eccentricities
- informs following dynamical criteria to be made for in situ formations (further supported by Leamen et al): ○ highly bound orbits (apo<3.5 kpc)
    - ○ z_max < 5 kpc ○ circ > 0.5
- some (~10) GCs are excluded from this given they are on the very edge of most of these criteria

- remaining GCs are divided into association with various merger events and analysed to find common traits that grouped them to said merger events ○ E
    - ○ L_z (angular momentum in z direction) ○ L_perp (angular perpendicular momentum)
- Sagittarius dwarf galaxy (~6 GCs) ○ 3700 < L_perp < 6200 km/s kpc ○ 0 < L_z < 3000 km/s kps
- Helmi streams (~7 GCs) ○ 350 < L_perp < 3000 km/s kpc ○ 1000 < L_z < 3200 km/s kps
    - ○ E (energy, found from integration? need to confirm but I remember checking and im pretty sure we can do this with the info we have) < 1*10^5
- Gaia Enceladus
    - ○ L_perp < 3500 km/s kpc ○ -800 < L_z < 620 km/s kps ○ (-1.86 < E < -0.96 ) *10^-5 km^2/s^2
    - ○ apo < 25 kpc
    - ○ strong AMR relation between these GCs as well

accreted vs in situ milky way globular clusters (Forbes et al)

- stars formed by accreted stellar haloes (which are commonly associated with accreted GCs and typically a product of mergers) are largely formed around 9 Gyrs ago, compared to in situ GCs which are formed roughly 11.5 Gyrs ago

- common high quality indication of detecting accretion is recording integrals of motion (like done before with the angular momentum but expanded upon a bit more here) which are typically conserved from their original stellar population post merger, setting it apart from surroundings
- other indications are ○ Oosterhoft type
    - unusualy element/metallicity rations that set it apart from surroundings ○ ellipticity ○ luminosity ○ size
    - horizontal branch kinematics
    - AMR (dwarf galaxies have very distinctive AMR relations making this one in particular useful)
    - velocity dispersion
- phase space can also be used to place stellar populations in space and connect them with merger events ○ done with Sagittarius and Canis Major dwarf galaxies
    - these together had a very similar AMR spread and curve on the graph ○ have been forming stars for roughly 10 Gyrs ○ both have similar luminosities/mass
- horizontal and extended horizontal branches on AMRs are useful but relatively less valid indications of accretion, extended horiztonal branches slightly more so but used more as method of confirmation rather than first steps
- retrograde motion is strong indication of GC being formed by satellite that had a motion retrograde to Milky way ○ z component of angular momentum was used to indicate this
- older GCs that were formed on timescale of <1Gyr is typically an indication of in situ formation

- overall this paper followed more or less the same steps as the previous one to identify GCs but just had some extra information that helped with my understanding a bit more + had good ideas for some graphs we could present

The chemo-dynamical groups of galactic globular clusters:

https://arxiv.org/abs/2202.00591

**this one was slightly more experimental/nebulous in its methods, a lot of it was visual pattern recognition rather than more quantitative distinctions, but I think it could be a useful/interesting method of confirming our preliminary findings in the video that we can touch on briefly**

- reiterates usefulness of integrals of motion, stellar age and chemical information that can be used to categorise and reconstruct GC history
- metal poor GCs are typically located further away from Milky way nucleus (which are typically where in situ GCs are found) while more metal rich GCs are located closer to the bulge (typically where accreted GCs are found) - most prominent accretion events:
    - Sagittarius ○ GES debris ○ Helmi streams ○ Sequoia
- correlation between total mass of the GC and the ancestor galaxy halo mass can be used to link GCs with ancestor galaxies and therefore indicate accretion
- dynamic action and AMR can also be used to do thos ○ angular momentum L_z ○ radial action J_r ○ vertical action J_z
- general process involved plotting all data of globular clusters (see paper for exact graphs). once this was done, graphs of above dynamic properties are used to identify distinct groups between overall spread, which are then applied to other dynamic graphs to gradually distinguish between all the various groups, which correspond to progenitor stellar populations/satellites via process of elimination
- estimation of these groups are then applied to AMR and line of best fits, which allows metallicity enrichment and formulation time to be found from line of best fit function, the values of which can be used to infer general accretion based on young-high metallicity relation

bifurcated AMR of milky way globular clusters Leamen

et al

http://ui.adsabs.harvard.edu/abs/2013MNRAS.436..1

22L/abstract

- $\Delta V_{TO}^{HB}$ (which is method from original paper Vandenberg et al) uses magnitude difference between horizontal branch and turnoff luminisoty as primary age

constraint ○ benefit of this is impact of associated metallicity uncertainties are minimised (good to metion in video)

- graph appears to be absolute vs difference between 2 relative magnitudes - AMR of milky way is bifurcated at roughly -1.8 Fe/H - use of phase space:
- consideration of GCs with respect to their host environment of milky way be identifiying them as either disk of halo populations (accreted GCs are more commonly located in halo while in situ located in disk)
  - halo associated with radius of 150 kpc (in terms of galaxy position)
  - disk associated with rotational velocity/velocity dispersion from established literature values (venn et all, 2004, table 3) - plotting galactocentric velocity against apocentric distance shows visual distinctions between disk and halo populations

- mass metallicity relations can be used to connect metallicity of GCs in halo with dark matter haloes of satellite galaxies such as SMC, LMC, Sagittarius, WLM
- plotting various properties against progenitor galaxy shows clear separation between milkly wayGCs and accreted GCs (figure 8) ○ this is again more of a confirmation rather than an actual method of distinction but good set of plots to have in video
  - log (M_GC) vs log (M_(*, progenitor))
  - v_esc vs log (M_(*, progenitor))
  - log (velocity dispersion) vs log (M_(*, progenitor))
  - r_per vs log (M_(*, progenitor))
  - r_apo vs log (M_(*, progenitor))